

Mojtaba Jafarian Abyaneh

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SUMMARY

PhD Candidate and Graduate Research Assistant at Florida Atlantic University specializing in AI and Machine Learning for transportation engineering. Builds Transformer models for Multi-Agent trajectory prediction from LiDAR, spatial graph networks for mobility flow, and city-scale CFD for hurricane pedestrian wind. Research funded by the NSF and NIH. First-author in *IEEE OJ-ITS*, with a second paper in revision for *IEEE T-ITS*.

EDUCATION

Florida Atlantic University <i>Ph.D. in Transportation and Environmental Engineering</i> GPA: 4.0/4.0 • <i>Research Focus: Deep learning for trajectory prediction and spatiotemporal mobility modeling</i> • <i>Advisor: Dr. Jinwoo Jang</i>	Boca Raton, FL 2024 – Present
University of New Brunswick <i>M.Sc. in Civil Engineering</i> • <i>Thesis: Vibration testing of cracked reinforced concrete beams under sustained load</i>	Fredericton, NB, Canada 2018 – 2020
Iran University of Science and Technology <i>M.Sc. in Civil/Earthquake Engineering</i> • <i>Thesis: Detection and estimation of structural damage using finite element model updating by means of damage-sensitive dynamic parameters</i>	Tehran, Iran 2014 – 2017
Iran University of Science and Technology <i>B.Sc. in Civil Engineering</i>	Tehran, Iran 2010 – 2014

RESEARCH INTERESTS

- Deep Learning for Intelligent Transportation Systems
- Multi-Agent Trajectory Prediction
- Spatial Graph Neural Networks
- Computer Vision & LiDAR Data Processing
- Urban Transportation Safety Modeling
- Urban Digital Twins & Computational Simulation
- Spatiotemporal Flow Prediction

EXPERIENCE

Doctoral Research Assistant Florida Atlantic University (I-SENSE Lab) • Developed SIPT for Multi-Agent trajectory prediction: ADE 0.76 m, 26.2% better than Social LSTM and 49% lower collision rate , on 64,000+ COSMOS trajectories; in revision for <i>IEEE T-ITS</i> . • Developed Transformer model for real-world LiDAR-based trajectory prediction: 15.24%/13.88% ADE improvement over LSTM (single-/Multi-Agent) on 12,390+ trajectories at a West Palm Beach intersection; published in <i>IEEE OJ-ITS</i> , Vol. 7 (2026). • Managing the NIH longitudinal dataset (R01AG068472): multi-sensor data and cognitive assessments from 750 older drivers, 12 waves over 3 years. • Co-mentoring two NSF CS3 REU students on GPS-based driver behavior analysis (road network enrichment, FMM map matching). • Running OpenFOAM CFD on HPC: hurricane pedestrian wind hazard for downtown Miami.	Aug. 2024 – Present Boca Raton, FL
Graduate Research Mentor Vertically Integrated Projects (VIP) Program, Florida Atlantic University • Overseeing 12 undergraduates in two VIP groups on AI smart-sensor applications for coastal resilience — computer vision, sentiment analysis, sensor analytics.	Jan. 2026 – Present Boca Raton, FL
Guest Lecturer Florida Atlantic University • CES 3102C Analysis of Structures : taught shear force and bending moment diagrams, with my <u>Interactive 2D Beam Solver</u> .	Sep. 2026 Boca Raton, FL

Data Analyst

Freelance

Nov. 2022 – Jul. 2024

Fredericton, NB, Canada

- Analyzed import/export data for trade compliance, tariff classification and logistics optimization.
- Built import-volume forecasting models and Tableau dashboards for shipment performance, cost trends and supplier metrics.

Graduate Research Assistant

University of New Brunswick

Sep. 2018 – Oct. 2020

Fredericton, NB, Canada

- Modeled reinforced concrete beams under sustained/cyclic loading in ABAQUS (concrete damage plasticity model); conducted vibration testing with accelerometers.
- Evaluated long-term deflection, natural frequencies, and mode shapes via ACI 318-19 / CSA A23.3 effective moment of inertia formulations.

Graduate Teaching Assistant

University of New Brunswick

Jan. 2019 – Apr. 2020

Fredericton, NB, Canada

- **Mechanics of Materials Experimental Lab** (Jan. – Apr. 2019, Sep. – Dec. 2019): Supervised undergraduates in material testing (tensile, compression, torsion), ensuring lab safety and accurate data collection.
- **Statics Computational Lab** (Jan. – Apr. 2020): Taught computational methods and statics, including force analysis, moment calculations, and structural equilibrium concepts.

Graduate Research Assistant

Iran University of Science and Technology

Sep. 2014 – Apr. 2017

Tehran, Iran

- Structural health monitoring and damage detection via optimization algorithms (grey wolf, genetic) for civil infrastructure.

PUBLICATIONS

1. **Jafarian Abyaneh, M.** and Jang, J. (2026) "Transformer-Based Trajectory Prediction Using LiDAR Data for Situational Awareness in Complex Urban Environments." *IEEE Open Journal of Intelligent Transportation Systems*, Vol. 7, pp. 16–28. [doi:10.1109/OJITS.2025.3640002](https://doi.org/10.1109/OJITS.2025.3640002)
2. Zare Hosseinzadeh, A., Ghodrati Amiri, G., **Jafarian Abyaneh, M.**, Seyed Razzaghi, S. A., and Ghadimi Hamzehkolaei, A. (2020) "Baseline updating method for structural damage identification using modal residual force and grey wolf optimization." *Engineering Optimization*, Vol. 52, No. 4, pp. 549–566. [doi:10.1080/0305215X.2019.1593400](https://doi.org/10.1080/0305215X.2019.1593400)
3. **Jafarian Abyaneh, M.** (2020) "Vibration testing of cracked reinforced concrete beams under sustained load." Master's thesis, University of New Brunswick.
4. Ghodrati Amiri, G., **Jafarian Abyaneh, M.**, and Zare Hosseinzadeh, A. (2016) "Model Updating-Based Approach for Damage Prognosis in Frames via Modal Residual Force." *International Journal of Civil, Environmental, Structural, Construction and Architectural Engineering*, Vol. 10, No. 8. [doi:10.5281/zenodo.1125800](https://doi.org/10.5281/zenodo.1125800)
5. Ghodrati Amiri, G., Zare Hosseinzadeh, A., **Jafarian Abyaneh, M.** (2015) "A new two-stage method for damage identification in linear-shaped structures via Grey System Theory and optimization algorithm." *Journal of Rehabilitation in Civil Engineering*, Vol. 3, No. 2, pp. 45–58. [doi:10.22075/jrce.2016.369](https://doi.org/10.22075/jrce.2016.369)

WORKING PAPERS & UNDER REVIEW

1. **Jafarian Abyaneh, M.**, Jang, J., Smyth, A. W., and Turkcan, M. K. (2026) "Spatial Interaction Pooling Transformer (SIPT) for Multi-Agent Trajectory Prediction in Urban Streetscapes." *IEEE Transactions on Intelligent Transportation Systems*. [In Revision]
2. **Jafarian Abyaneh, M.**, Jang, J., and Kaiser, E. I. (2025) "Spatial Graph Convolutional Network for Predicting Bike-Sharing Origin-Destination Spatiotemporal Flows." SSRN Working Paper No. 5801889. [doi:10.2139/ssrn.5801889](https://doi.org/10.2139/ssrn.5801889) [Preprint]
3. **Jafarian Abyaneh, M.**, Jang, J., Smith, D. K., and Lalla, J. (2026) "The Impact of Building Geometry Fidelity on Hurricane Pedestrian Wind Hazard Across Inflow Conditions." [In Preparation]

CONFERENCE PRESENTATIONS

1. **Jafarian Abyaneh, M.** and Jang, J. (Jan. 2027) "Geometry-Aware Transformer for Multi-Agent Trajectory Forecasting at Urban Intersections." *106th Transportation Research Board (TRB) Annual Meeting*, Washington, DC. Poster presentation. [Accepted]
2. **Jafarian Abyaneh, M.**, Jang, J., and Kaiser, E. I. (Jan. 2026) "Spatial Graph Convolutional Network for Predicting Bike-Sharing Origin-Destination Spatiotemporal Flows." *105th Transportation Research Board (TRB) Annual Meeting*, Washington, DC. Poster presentation.

WORKSHOP PRESENTATIONS

1. **Jafarian Abyaneh, M.** and Jang, J. (May 2026) “Predicting the Crowd: Spatially-Informed Transformers for Urban Traffic Agents.” *CS3 NSF Renewal Site Visit, Columbia Engineering Innovation Hub — CS3 HQ, New York, NY*. Poster presentation.
2. Smith, D. K., **Jafarian Abyaneh, M.**, and Jang, J. (May 2026) “Voxelized High-Fidelity 3D Models for City-scale Wind Simulations.” *CS3 NSF Renewal Site Visit, Columbia Engineering Innovation Hub — CS3 HQ, New York, NY*. Poster presentation.
3. **Jafarian Abyaneh, M.** and Jang, J. (Mar. 2026) “Context-Aware Deep Learning for Predicting Pedestrian and Vehicle Movement Patterns in Urban Traffic Scenarios.” *8th Annual Celebration of Academic Excellence, College of Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL*. Poster presentation.
4. Smith, D. K., **Jafarian Abyaneh, M.**, and Jang, J. (Mar. 2026) “Testing the Accuracy of Urban 3D Models With Airflow Simulations.” *Florida Undergraduate Research Conference (FURC), University of North Florida, Jacksonville, FL*. Poster presentation.
5. **Jafarian Abyaneh, M.** and Jang, J. (Feb. 2026) “Spatial Interaction Pooling Transformer (SIPT) for Multi-Agent Trajectory Prediction.” *CS3 Innovation Summit, Columbia University, New York, NY*. **Perfect Pitch Competition** (90-second research presentation) and **Poster Presentation**.
6. Smith, D. K., Lalla, J., **Jafarian Abyaneh, M.**, and Jang, J. (Feb. 2026) “Automated Voxelization Pipeline for High-Fidelity Urban Digital Twins.” *CS3 Innovation Summit, Columbia University, New York, NY*. Poster presentation.

RESEARCH FUNDING

- **NSF** Engineering Research Center for Smart Streetscapes (CS3), Award No. EEC-2133516 — Graduate Researcher
- **NIH** In-Vehicle Sensors to Detect Cognitive Change in Older Drivers, Award No. R01AG068472 — Graduate Researcher
- **FAU** Vertically Integrated Projects (VIP): Smart Sensors and AI for Coastal Destination Resilience — Graduate Research Mentor

UNDERGRADUATE RESEARCH MENTORING

NSF CS3 (EEC-2133516) — Digital Twins for Smart Streetscapes, FAU Jan. 2026 – Present

- **Dana K. Smith** (Civil Eng., FAU) — voxelization & CFD-ready digital twin pipelines
- **Jonathan Lalla** (CS, FAU) — voxelization & CFD-ready digital twin pipelines
- **Cyrus Khan** (CS, UC Riverside; Summer REU) — GPS-to-behavior-metric pipeline, Fast Map Matching (FMM)
- **Vedant Sundriyal** (CS, Lehman College; Summer REU) — QuASAR: watertight 3D city reconstruction ([GitHub](#))

FAU VIP Program — Smart Sensors and AI for Coastal Destination Resilience Jan. 2026 – Present

- **Environment & Sustainability:** Dana K. Smith, Iris York, Felicia Phan, James Kane, Dana Urbano Prieto, Sammie Chamberlain (FAU)
- **Safety & Security:** Jonathan Lalla, Mykyta Garkusha, Mohammed Mered, Howard Lowe, Nicolette Madrazo (FAU)

HONORS AND AWARDS

- ASCE Palm Beach Branch Scholarship (Apr. 2026) — \$1,000 competitive award
- New Brunswick Innovation Foundation (NBIF) Graduate Award (2018–2019)
- Esterbauer Civil Engineering Graduate Award, University of New Brunswick (2018–2020)

PROFESSIONAL AFFILIATIONS

- Institute of Electrical and Electronics Engineers (IEEE), Member
- American Society of Civil Engineers (ASCE), Member

TECHNICAL SKILLS

- **Programming & Tools:** Python, MATLAB, LaTeX, Typst, Git, Tableau
- **AI & Machine Learning:** PyTorch, TensorFlow, Keras
- **Data Science:** Pandas, NumPy, Scikit-learn, GeoPandas, ArcGIS
- **Engineering Software:** ABAQUS, OpenFOAM (HPC), ParaView, SAP2000, ETABS, SAFE, Synchro, Civil 3D, MicroStation, AutoCAD